

PROBLEM SET 2

ECON301—ECONOMETRICS I

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1. Suppose that we have n independent observations (X_i, Y_i) of the random vector (X, Y) . Under Assumptions 1 to 4 find and justify the moments below. Try to make use as much as possible of the Theorems seen in the lectures.
 - i. $E(\bar{Y}|X's)$
 - ii. $E(\hat{\beta}_0 + \hat{\beta}_1 X_1|X's)$
 - iii. $\text{Var}(\hat{\beta}_0 + \hat{\beta}_1 X_1|X's)$
 - iv. $\text{Var}(\bar{Y} - \beta_1 \bar{X}|X's)$
 - v. $\text{Cov}(\bar{Y}, \hat{\beta}_1|X's)$
 - vi. $E(\bar{Y} - \hat{\beta}_1 X_1|X's)$
 - vii. $\text{Var}(\bar{Y} - \hat{\beta}_1 X_1|X's)$
2. Under Assumptions 1 to 4 derive $E(\hat{\beta}_0|X's)$. *Hint: make use of $E(\hat{\beta}_1|X's)$.*
3. Under Assumptions 1 to 4 derive the full matrix $\text{Var}(\underline{\hat{\beta}}|X's)$. *Hint: make use of the expression for $\text{Var}(\hat{\beta}_1|X's)$ and the result from 2.f above.*
4. Suppose that we have three independent observations (X_i, Y_i) of the random vector (X, Y) . Assume that $E(Y_i|X_i) = 2X_i + 3$, and $\text{Var}(Y_i|X_i) = 3$.
 - i. If $X_1 = 1$, $X_2 = 3$, and $X_3 = 6.5$, is assumption 3 fulfilled?
 - ii. Find the numeric values of $E(\bar{Y} - \hat{\beta}_1 X_1|X's)$, and $\text{Var}(\hat{\beta}_0 + \hat{\beta}_1 X_1|X's)$.
 - iii. Indicate the numeric values of $\text{Var}(\underline{\hat{\beta}}|X's)$ (the whole matrix).
5. Suppose that we have n independent observations (X_i, Y_i) of the random vector (X, Y) with pdf $f(x, y)$. Show that
 - i. $f(y_1, \dots, y_n|x_1, \dots, x_n) = \prod_{i=1}^n f(y_i|x_i)$. This is called *conditional independence*.
 - ii. $f(y_i|x_1, \dots, x_n) = f(y_i|x_i)$ for $i = 1, \dots, n$.
 - iii. $f(y_i, y_j|x_1, \dots, x_n) = f(y_i|x_i)f(y_j|x_j)$ for any $i \neq j$.
 - iv. If $\text{Var}(Y_1) < \infty$ then $\text{Cov}(Y_i, Y_j|X_1, \dots, X_n) = 1_{\{0\}}(i - j)\text{Var}(Y_1|X_1)$.
6. Suppose that we have n independent observations (X_i, Y_i) of the random vector (X, Y) with pdf $f(x, y) = (x + y)1_{[0,1]}(x)1_{[0,1]}(y)$. Let $(\hat{\beta}_0, \hat{\beta}_1)$ be the LSE. Find $E(\hat{\beta}_1|X's)$ and compute its numerical value when $X_1 = 1$, $X_2 = 3$, and $X_3 = 6.5$.
7. By definition, a Student t random variable with k degrees of freedom has the form

$$t = \frac{Z}{\sqrt{V/k}}$$

where $Z \sim N(0, 1)$, $V \sim \chi^2(k)$ and Z and V are independent. Suppose assumptions 1 to 5 from the topic 2 hold. Assuming that $(\hat{\beta}_0, \hat{\beta}_1)$ and $\hat{\sigma}^2$ are independent, show that

$$\frac{\hat{\beta}_i - \beta_i}{\text{SE}(\hat{\beta}_i|X's)} | X's \sim t(n - 2)$$

8. Use the result from problem 7 to construct a $(1 - \alpha)\%$ confidence interval for β_1 . Find the numerical value of the interval if $\alpha = 0.05$, $n = 3$, $X_1 = 1$, $X_2 = 3$, $X_3 = 6.5$, and $\sum(\hat{\epsilon}_i)^2 = 8.6$.